University of Idaho
Department of Computer Science
Coeur d'Alene

Lecture #6

Convolution and Kernels

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- Affine Transformation, pronounced *a-fine*
- $T = M \begin{bmatrix} x & y & 1 \end{bmatrix}^T$
  - Translation
  - Rotation
  - Shear
  - Scaling
- Other forms of transformation:
  - Perspective
  - Color/Brightness

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What is Convolution?

- Convolution has a mathematical definition
- Computer Science adapts this to various use cases
  - Change parameters/behavior based on input data
  - Filters
  - Convolutional Neural Networks

Mathworks: Convolution | 3Blue1Brown: What is a convolution? | 3Blue1Brown: What is a convolution?

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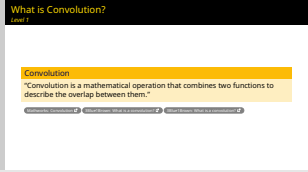
What is Convolution?

Level 1

Convolution

“Convolution is a mathematical operation that combines two functions to describe the overlap between them.”

Mathworks: Convolution | 3Blue1Brown: What is a convolution? | 3Blue1Brown: What is a convolution?



1. For example: given two dice whose probability of rolling various numbers varies(rolling a one on A happens 11%, vs 5% on B), how do we calculate the probability of each possible sum produced by the dice(2,3,4,5,6,7,8,9,10,11,12)?

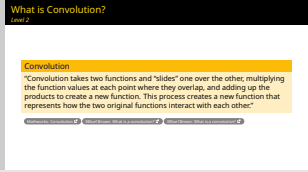
What is Convolution?

Level 2

Convolution

“Convolution takes two functions and “slides” one over the other, multiplying the function values at each point where they overlap, and adding up the products to create a new function. This process creates a new function that represents how the two original functions interact with each other.”

Mathworks: Convolution | 3Blue1Brown: What is a convolution? | 3Blue1Brown: What is a convolution?



1. For example: given two dice whose probability of rolling various numbers varies(rolling a one on A happens 11%, vs 5% on B), how do we calculate the probability of rolling a 5(described by overlap)? Or any other combination produced by the dice(2,3,4,5,6,7,8,9,10,11,12)?
2. “You can imagine convolution as the mixing of information. Imagine two buckets full of information which are poured into one single bucket and then mixed according to a specific rule. Each bucket of information has its own recipe, which describes how the information in one bucket mixes with the other. So convolution is an orderly procedure where two sources of information are intertwined.” Tim Dettmers, 2015

What is Convolution?

Level 3: Formal Definition

Convolution

The integral of the product of two functions after one is reflected about the y-axis, $g(-\tau)$, and shifted by t.

$$(f * g)(t) \approx^{def} \int_{-\infty}^{\infty} f(\tau)g(t-\tau)dr$$

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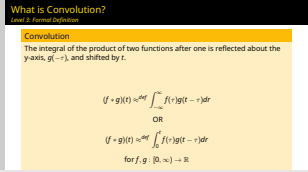
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OR

$$(f * g)(t) \approx^{def} \int_{0}^{t} f(\tau)g(t-\tau)dr$$

$$\text{for } f, g : [0, \infty) \to \mathbb{R}$$



1. A limit from $[0, t]$ makes a lot more sense for cases such as pixel values which are clipped to range $[0, 255]$.

Convolution

for CS...

- Pair-wise multiplication(dot product) of two matrices (could be reformatted as 1D arrays)
  - Each matrix represents a function/distribution of data
  - Smaller matrix represents “weights” to apply to input data
- Summing the resulting values to produce *one* output value

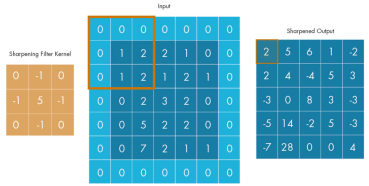


Figure: Convolution with 2 Matrices

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Figure: Convolution with 2 Matrices

1. If this page only contains a static image, that means your pdf viewer does not support javascript(try acrobat). Not a problem, see the “Animation” slide below for a link to the original video.

What is a Kernel?

- The matrix moved across our data set or alternate function.
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- The matrix moved across our data set or alternate function.
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1. This makes kernels the flipped/translated function in our formal definition of convolution.
2. Generally want kernel to have an odd dimension to reduce dimensionality of output and coalesce data to a set of “features”. The single output value represents a feature.
3. In mathematical definition, the convolution of two functions *is larger* than the input because partial overlaps are allowed. So more possible outputs than inputs.
4. Note that the odd dimension requirement also means a kernel less than 3x3 is relatively meaningless, or is a simple scaling of image values(1x1).

Kernels, Convolution, and Machine Vision

- Convolution is used in several ways in MV
- Convolution produces a “new image” or an alternate interpretation of the image

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Animation

Convolution Animation(CS/Kernel style)

Animation

Convolution by Hand

$$I = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 2 & 0 & 0 \\ 0 & 2 & 0 & 3 & 3 & 0 & 0 \\ 0 & 0 & 2 & 3 & 3 & 1 & 0 \\ 0 & 1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} K = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

What is the output?

$$\begin{bmatrix} a_{00} & a_{01} & a_{02} \\ a_{10} & a_{11} & a_{12} \\ a_{20} & a_{21} & a_{22} \end{bmatrix}$$

1.

$$Output = \begin{bmatrix} 0 & 3 & -3 \\ 1 & -3 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

Formal Explanation

From OpenCV Docs...

From [1]

1. Place the kernel anchor on top of a determined pixel, with the rest of the kernel overlaying the corresponding local pixels in the image.
2. Multiply the kernel coefficients by the corresponding image pixel values and sum the result.
3. Place the result to the location of the anchor in the input image.
4. Repeat the process for all pixels by scanning the kernel over the entire image.

$$H(x,y) = \sum_{i=0}^{M_i-1} \sum_{j=0}^{M_j-1} I(x+i-a_i, y+j-a_j)K(i,j)$$

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$H(x,y) = \sum_{i=0}^{M_i-1} \sum_{j=0}^{M_j-1} I(x+i-a_i, y+j-a_j)K(i,j)$

1. In the algorithm, point (x,y) corresponds to the pixel in image coordinates, which the *anchor* of the kernel is placed above. This point is passed to H.
2. $I(x+i-a_i, y+j-a_j)$ is the image pixel value, corresponding to Kernel coefficient $K(i,j)$.

Convolution and Kernels

- Note
  - sometimes images are “padded” to match dimensions with kernel
  - often want output dimension to be smaller than input
  - how many pixels the kernel moves is referred to as “stride”.

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1. Stride is a term from CNN API implementations, and successive convolutional layers also produce lower dimensional outputs as they learn a set of influential “features”.

Image Size

- If we don’t want image to shrink → add padding
- `cv.copyMakeBorder()`

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Popular Uses for Filters

- Averaging
- Blurring/Smoothing
- Noise
- Convolutional Neural Networks

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Filters in OpenCV

cv.filter2D()

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cv.filter2D(src,
            ddepth, # desired depth(data precision)
            kernel, # a single-channel floating point matrix
            anchor=Point(-1,-1), # center point of kernel
            delta=0, # optional value added to filtered pixels
            borderType=BORDER_DEFAULT # extrapolation method
)
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1. Set ddepth=-1*topreservedataprecisionofinputimage*

Gaussian Kernel

- Gaussian distribution = Normal Distribution
- Carl Freidrich Gauss [2]
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- Useful for blurring, smoothing [3]

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1. Why use the gaussian? Why does normal distribution matter?
2. Gaussian distribution gives greatest weight to center/anchor of kernel. Thus output provides a blurring effect, but retains more information from center pixel than those farther from center.

Other Uses for Kernels

- Image Gradients!

Other Uses for Kernels
- Image Gradients!

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[1] “OpenCV: Making your own linear filters!” (), [Online]. Available: https://docs.opencv.org/4.x/d4/dbd/tutorial_filter_2d.html (visited on 08/27/2024).

[2] “Carl Friedrich Gauss | Biography, Discoveries, & Facts | Britannica,” (Aug. 13, 2024), [Online]. Available: https://www.britannica.com/biography/Carl-Friedrich-Gauss (visited on 08/27/2024).

[3] “Spatial Filters - Gaussian Smoothing,” (), [Online]. Available: https://homepages.inf.ed.ac.uk/rbf/HIPR2/gsmooth.htm (visited on 08/27/2024).

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[1] “OpenCV: Making your own linear filters!” (), [Online]. Available: https://docs.opencv.org/4.x/d4/dbd/tutorial_filter_2d.html (visited on 08/27/2024).

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